

Team B: Guardians

Project Criteria : Process Automation

Project Name: Student Attendance Management

• **Project Overview**

The Attendance Automation System is a backend application that automates the process of managing daily and weekly student attendance.

It allows the system to prepare a daily attendance file (`dialyAttendance.csv`) by adding a fresh date column, programmatically launching it in MS Excel for the faculty, and pausing execution until it is closed.

The system then extracts only that specific date's column and appends it horizontally (side-by-side) into a master file (`weeklyAttendance.csv`) based on a unique student identifier (Roll Number), followed by an automated weekly email report.

• **Project Objectives**

1. Automate the side-by-side syncing of daily attendance data into a master ledger.
2. Reduce manual effort in copying data between multiple Excel sheets.
3. Improve data integrity by ensuring student records align perfectly across files.
4. Support automated email reporting at designated cycle intervals.
5. Maintain clean operational logs and handle file locks gracefully.

• **Functional Requirements**

1. The system should read `dialyAttendance.csv` and automatically append the current date (YYYY-MM-DD) as a new column title before the user interacts with it.
2. The system should prevent duplicate date columns if the program is executed multiple times on the same calendar day.
3. The system should automatically open the daily CSV file in the OS default handler (MS Excel) and pause its background execution until the user saves and completely closes Excel.
4. The system should read the closed daily file and isolate only the newly added date column data.
5. The system should read `weeklyAttendance.csv` and merge today's column horizontally (side-by-side) by matching the unique student **Roll No** (Column 0).
6. The system should ensure that existing historical data and student names in the master file are never overwritten or altered during the merge.
7. The system should evaluate the current weekday and automatically trigger the email dispatch routine if it matches the designated target day (e.g., Monday for testing, Friday for production).
8. The system should securely connect to Gmail's SMTP server (Port 465) using an App Password configuration.

9. The system should automatically attach the updated `weeklyAttendance.csv` file and send the report to the designated stakeholder email address.
10. The system should detect and handle common runtime errors, such as a file being locked by another process (`PermissionError`) or missing paths, and display user-friendly logs instead of crashing.
11. The system must be built entirely using Python's native standard libraries without relying on any external packages (no `pip install`).

• **Acceptance Criteria**

1. Dynamic Column Preparation
2. Synchronous File Launch & Pause
3. Horizontal Data Merging by Primary Key (Roll No)
4. Weekday Evaluation & Automated Schedule
5. Secure SMTP Email and Attachment Transmission
6. Graceful Error and File-Lock Handling